SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

CO4_AT1 — RELATIONSHIP VISUALIZATION LAB

Course Code:	DSA06	**Course Title:**	Data Handling and Visualization
CO Assessed:	CO4 – Analysis (BL4)	**Assessment:**	Assessment Tool 1 – Relationship Visualization Lab
Weightage:	20% of CIA	**Total Marks:**	25 (5 Questions × 5 Marks)
CO4 Statement:	*Analyze relationships between variables using scatterplots, correlograms, PCA, bubble charts, and network graphs to construct and interpret appropriate visualizations.(BL4)*		
Student Name:	Sania Herbert	**Reg. No.:**	192324062
Section / Batch:	2023/AI&DS	**Date:**	23.08.26

Code of Conduct

I Sania Herbert(192324062) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Sania

Instructions

- This test contains 5 questions, each carrying 5 marks. Total: 25 Marks.
- No negative marking. Unanswered questions carry zero marks.
- No calculators or electronic devices permitted.
- Duration: 90 minutes.
- Graph paper or a ruler is recommended for accurate, to-scale plotting.
- Show all supporting calculations (e.g., variance explained, correlation strength, node degree) where relevant.
- Every chart must include a title, correctly labeled axes, and a legend where relevant.

Annexure A – RELATIONSHIP VISUALIZATION LAB

Rubrics for Calculation **ASSESSMENT RUBRIC**

AT1 — Relationship Visualization Lab

CO4 · BT4: Analyze ·

This rubric is used to assess student responses to the CO4-AT1 Relationship Visualization Lab problems, in which students construct and analyze scatterplots, correlograms, PCA outputs, bubble charts, or network graphs from given data.

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	**25%**	Demonstrates thorough understanding of the relevant relationship-visualization technique (scatterplot, correlogram, PCA, bubble chart, or network graph) and correctly applies it to the given data.	Good understanding of the relevant technique, with only minor gaps in application.	Basic understanding, but with some misconceptions about the correct technique or how to apply it.	Poor understanding of core concepts; applies a technique fundamentally unsuited to the data.
Explanation	**25%**	Provides detailed, well-reasoned analysis of the constructed visualization, explicitly tied to the specific values in the dataset.	Provides clear analysis with adequate reasoning tied to the data.	Analysis is present but generic or only loosely connected to the specific dataset given.	Little or no analytical reasoning provided; answers restate the data without interpretation.
Application	**20%**	Effectively applies construction and analysis techniques to produce a complete, accurate visualization	Applies techniques to produce a correct visualization, with only minor errors.	Limited application; the visualization partially represents the data but has notable errors.	Fails to apply appropriate techniques; the visualization does not correctly represent the data.

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
		correctly matched to the data.			
Organization	**15%**	The response is clearly structured, with construction and analysis presented in a logical sequence with coherent overall flow.	The response is organized, with only minor issues in flow or clarity.	Basic structure; construction and analysis are present but the response lacks clarity or sequence.	Poorly organized; the response is incoherent or key parts are left unaddressed.
Accuracy	**10%**	All parts of the answer — construction, calculations, and analysis — are completely accurate and well-suited to the given data.	Mostly accurate, with only minor omissions or a small error.	Partially correct; some elements are missing, mismatched to the data, or incorrect.	Largely incorrect or inappropriate, with major gaps across multiple parts.

Scoring Guide

For each criterion, award a performance level (1–4) based on the descriptors above. Multiply the level achieved (as a percentage of 4, i.e., Level 4 = 100%, Level 3 = 75%, Level 2 = 50%, Level 1 = 25%) by the criterion's weightage, then sum across all five criteria to obtain the total score out of 100%.

Faculty In-charge	Course Coordinator	Head of Department
Signature: _______________	Signature: _______________	Signature: _______________
Date: ___________________	Date: ___________________	Date: ___________________

Problem 5

Data: Screen time (hrs/day) and sleep hours for 8 teenagers: (2,8.5), (3,8), (4,7.5), (5,7), (6,6.5), (7,6),
(8,5.5), (9,5).

Task: Construct a scatterplot and identify the type of relationship shown.

A **scatterplot** is a graph used to show the relationship between two numerical variables. Each point represents one observation, where the **x-axis** represents screen time and the **y-axis** represents sleep hours.

In this data, as screen time increases from **2 hours to 9 hours per day**, sleep decreases from **8.5 hours to 5 hours**.

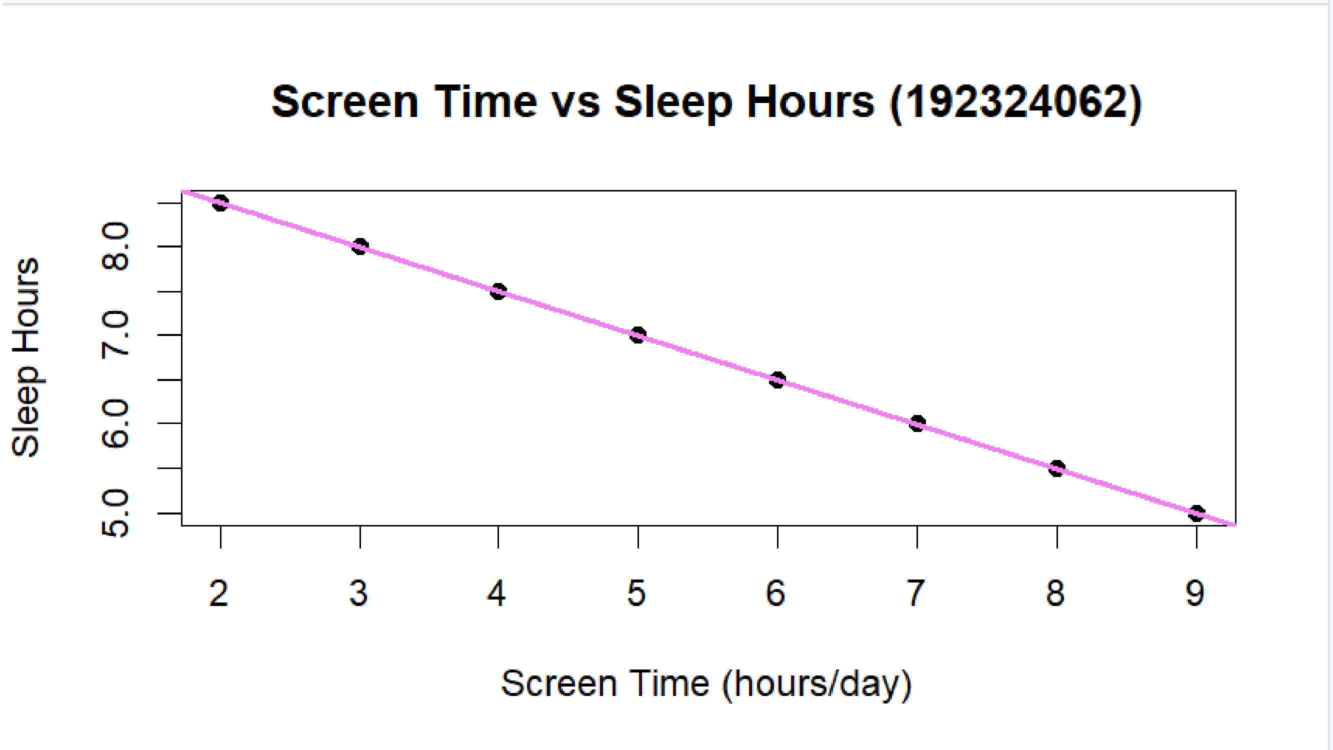
Therefore, the points in the scatterplot move **downward from left to right**, indicating a **negative linear relationship**.

```r
# Data
screen_time <- c(2, 3, 4, 5, 6, 7, 8, 9)

sleep_hours <- c(8.5, 8, 7.5, 7, 6.5, 6, 5.5, 5)

# Create scatterplot
plot(screen_time,
     sleep_hours,
     main = "Screen Time vs Sleep Hours (192324062)",
     xlab = "Screen Time (hours/day)",
     ylab = "Sleep Hours",
     pch = 19)

# Add a trend line
abline(lm(sleep_hours ~ screen_time),
       col = "violet",
       lwd = 2)
```

17:16 (Top Level) R Script

R 4.6.1 · ~/

```
> # Create scatterplot
> plot(screen_time,
+      sleep_hours,
+      main = "Screen Time vs Sleep Hours (192324062)",
+      xlab = "Screen Time (hours/day)",
+      ylab = "Sleep Hours",
+      pch = 19)
>
> # Add a trend line
> abline(lm(sleep_hours ~ screen_time),
+        col = "violet",
+        lwd = 2)
>
```



Interpretation

The scatterplot shows a **strong negative linear relationship** between screen time and sleep hours.

- As **screen time increases**, **sleep hours decrease**.
- For example, at **2 hours** of screen time, sleep is **8.5 hours**.
- At **9 hours** of screen time, sleep decreases to **5 hours**.
- The points form an almost straight downward line.

Conclusion: There is a **negative correlation** between screen time and sleep duration. Therefore, teenagers with higher daily screen time tend to get fewer hours of sleep.

Problem 6

Data: Correlation values for 3 variables: Height–Weight = 0.85, Height–Age = 0.15, Weight–Age = 0.30.

Task: Construct a correlogram (describe the shading/color for each cell) and identify the strongest pair.

A **correlogram** is a visual representation of the correlation matrix. The correlation coefficient ranges from **−1 to +1**.

- Values close to **+1** indicate a strong positive relationship.
- Values close to **0** indicate a weak or no linear relationship.
- In a typical correlogram, **darker shading represents stronger correlation**, while lighter shading represents weaker correlation.

Correlation Matrix

Height	Weight	Age
1.00	**0.85**	0.15
0.85	1.00	0.30
0.15	0.30	1.00

Shading / Color Description

- **Height–Height = 1.00:** Very dark/strong positive color because it is a perfect self-correlation.
- **Weight–Weight = 1.00:** Very dark/strong positive color.
- **Age–Age = 1.00:** Very dark/strong positive color.
- **Height–Weight = 0.85: Dark positive shading**, showing a strong positive correlation.
- **Height–Age = 0.15: Very light positive shading**, showing a weak relationship.
- **Weight–Age = 0.30: Light positive shading**, showing a weak-to-moderate relationship.

Strongest Pair

The strongest pair is:

Height–Weight = 0.85

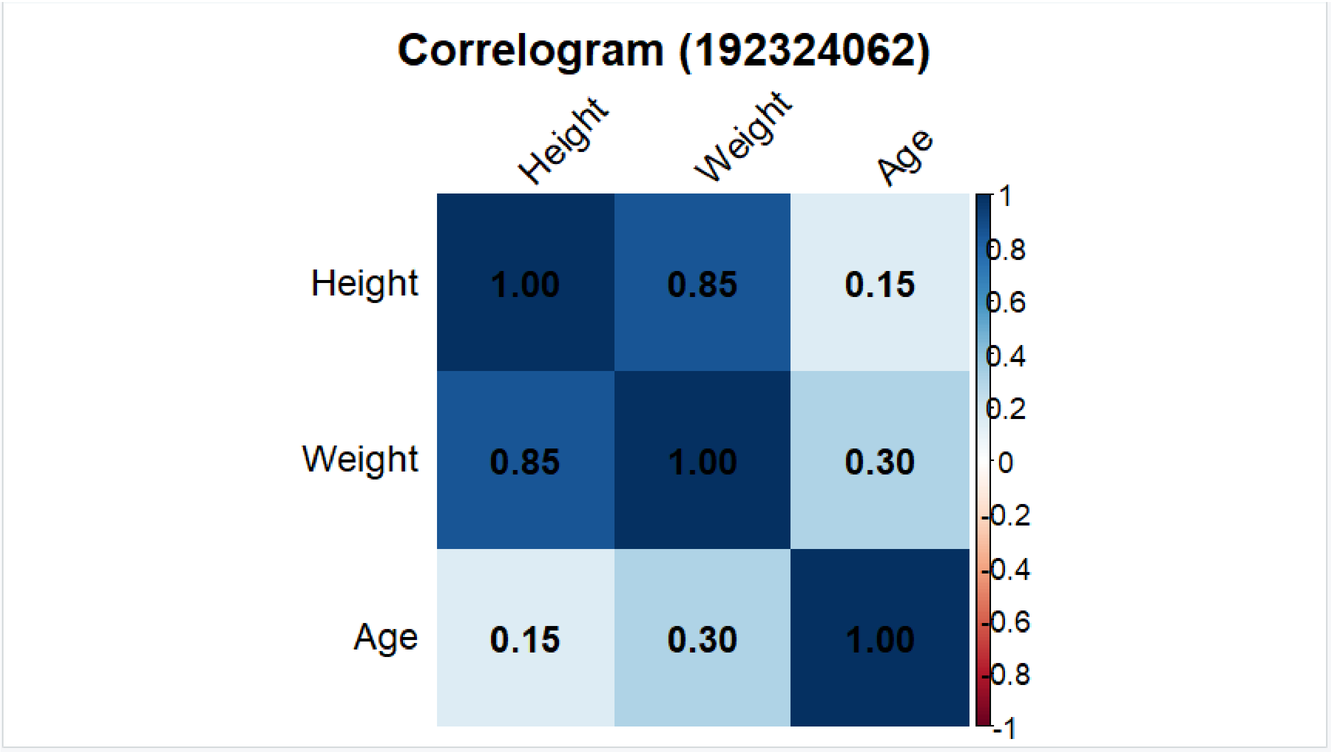
Therefore, **Height and Weight have the strongest positive correlation** among the three variable pairs.

```r
# Display correlation matrix
print(cor_matrix)

# Install corrplot if required
# install.packages("corrplot")

library(corrplot)

# Construct correlogram
corrplot(cor_matrix,
         method = "color",
         addCoef.col = "black",
         tl.col = "black",
         tl.srt = 45,
         title = "Correlogram",
         mar = c(0, 0, 2, 0))
```

23:30 (Top Level) R Script

R 4.6.1 · ~/

```
corrplot 0.95 loaded

>
> # Construct correlogram
> corrplot(cor_matrix,
+          method = "color",
+          addCoef.col = "black",
+          tl.col = "black",
+          tl.srt = 45,
+          title = "Correlogram",
+          mar = c(0, 0, 2, 0))
>
```



Problem 12

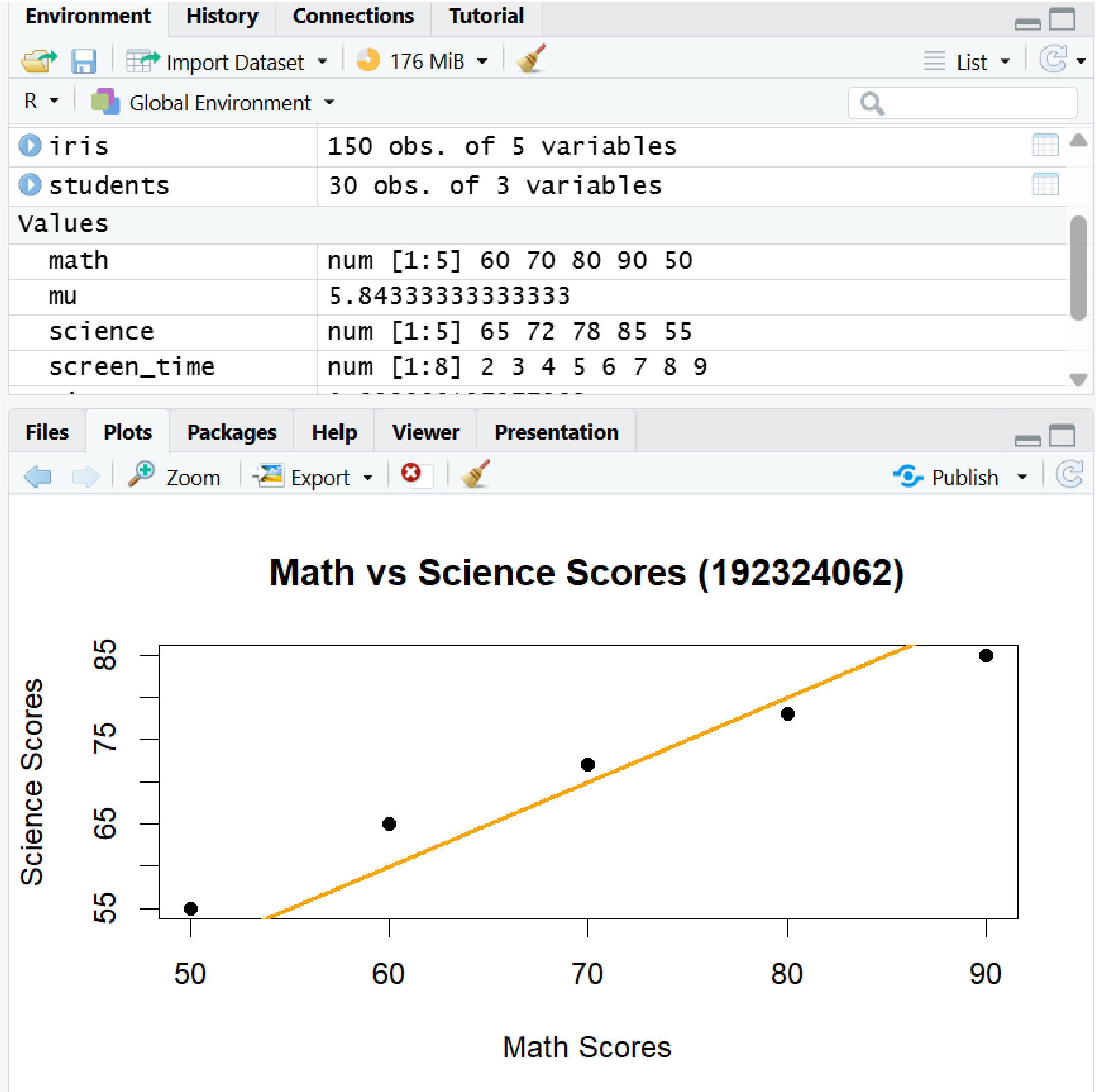
Data: Math and Science scores for 5 students: (60,65), (70,72), (80,78), (90,85), (50,55).

Task: Plot the scatter and explain whether PC1 would align closely with the diagonal of the plot, and why.

Student	Math	Science
1	60	65
2	70	72
3	80	78
4	90	85
5	50	55

Here, **Math scores** are plotted on the X-axis and **Science scores** on the Y-axis. The points show that students who score higher in Math generally also score higher in Science. Therefore, the two variables have a **strong positive relationship**. **PC1 (Principal Component 1)** is the direction that captures the maximum variation in the data. Since Math and Science scores increase together, the maximum variation is approximately along a **diagonal line from the lower-left to the upper-right**. Therefore, **PC1 would align closely with the diagonal of the scatter plot** because the two variables are strongly positively correlated.

```r
# Data
math <- c(60, 70, 80, 90, 50)
science <- c(65, 72, 78, 85, 55)

# Scatter plot
plot(math, science,
     main = "Math vs Science Scores (192324062)",
     xlab = "Math Scores",
     ylab = "Science Scores",
     pch = 19)

# Add diagonal reference line
abline(a = 0, b = 1, col = "orange", lwd = 2)
```



Problem 18

Data: 6 products' Price (X, $), Customer Rating (Y, /5), Units Sold (bubblesize):P1(20,4.5,500),P2(35,3.8,300),P3(15,4.2,700),P4(50,4.8,150),P5(25,4.0,600), P6(40,3.5,250).

Task: Construct a bubble chart and identify which product has the best combination of high rating and
high sales.

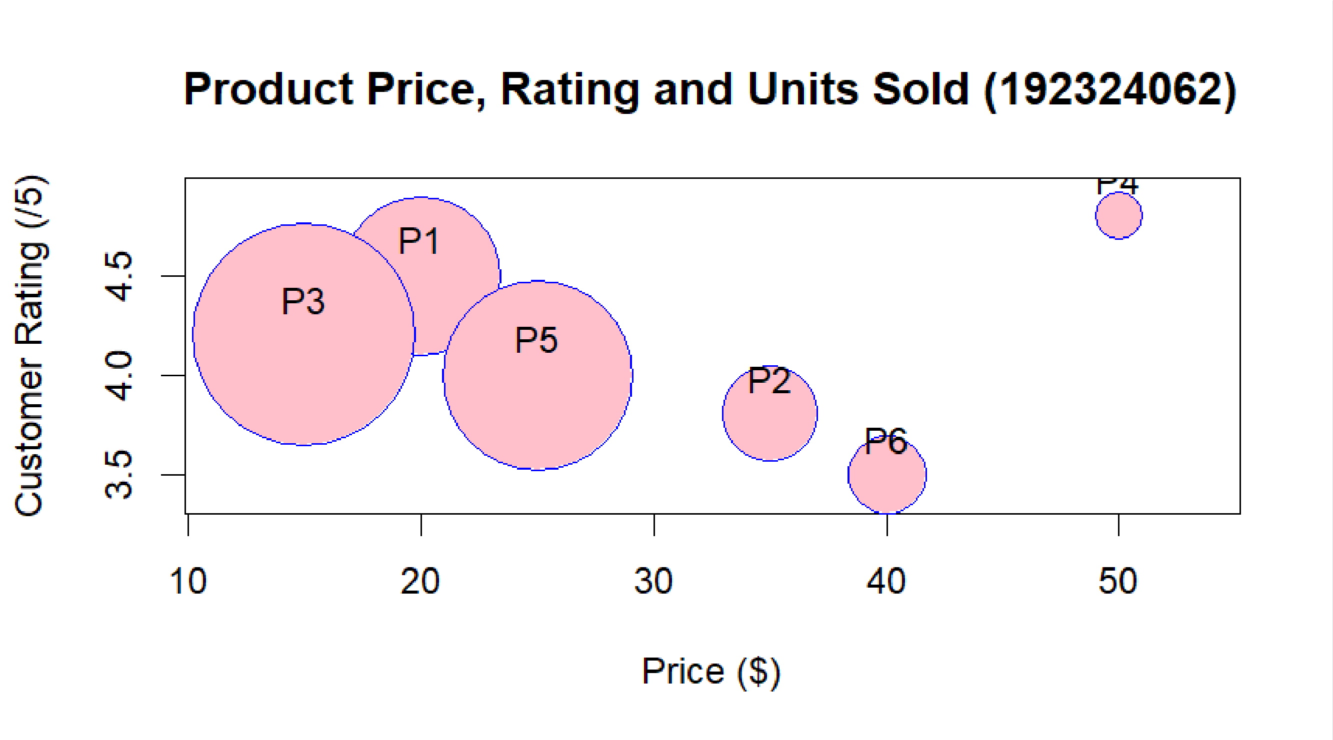
Product	Price ($)	Customer Rating (/5)	Units Sold
P1	20	4.5	500
P2	35	3.8	300
P3	15	4.2	700
P4	50	4.8	150
P5	25	4.0	600
P6	40	3.5	250

A **bubble chart** is an extension of a scatter plot that represents a **third numerical variable using the size of the bubbles**.

- **X-axis:** Price ($)
- **Y-axis:** Customer Rating
- **Bubble size:** Units Sold

A larger bubble indicates a higher number of units sold.

```
# Create data
product <- c("P1", "P2", "P3", "P4", "P5", "P6")
price <- c(20, 35, 15, 50, 25, 40)
rating <- c(4.5, 3.8, 4.2, 4.8, 4.0, 3.5)
units_sold <- c(500, 300, 700, 150, 600, 250)

# Bubble chart
symbols(price, rating,
        circles = units_sold,
        inches = 0.5,
        fg = "blue",
        bg = "pink",
        xlab = "Price ($)",
        ylab = "Customer Rating (/5)",
        main = "Product Price, Rating and Units Sold")

# Add product labels
```



Problem 24
Data: A collaboration network of 6 researchers: R1–R2, R1–R3, R2–R3, R4–R5, R5–R6.
Task: Draw the network graph and determine whether it forms one connected group or multiple
separate groups.

A **network graph** represents relationships between entities using:

- **Nodes (vertices):** Researchers.
- **Edges (links):** Collaborations between researchers.

Given collaborations:

- R1 – R2
- R1 – R3
- R2 – R3
- R4 – R5
- R5 – R6

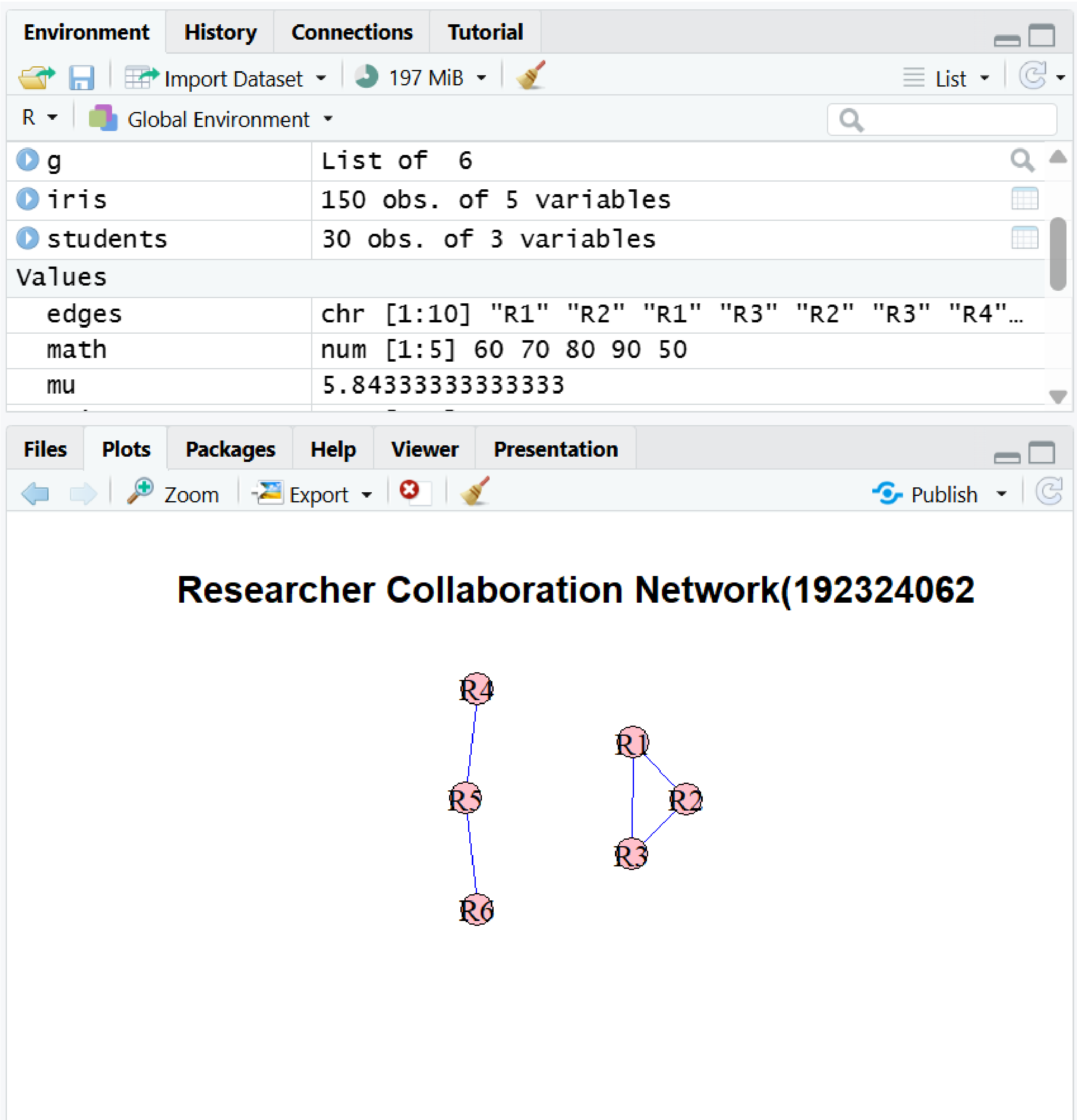
The network forms **two separate groups**:

1. **Group 1:** R1, R2, R3

2. **Group 2:** R4, R5, R6

There is **no edge connecting these two groups**. Therefore, the complete network is **not connected** and contains **2 connected components**.

```
  "R1", "R2",
  "R1", "R3",
  "R2", "R3",
  "R4", "R5",
  "R5", "R6"
)

# Create network graph
g <- graph(edges = edges, directed = FALSE)

# Draw network graph
plot(g,
     vertex.color = "pink",
     vertex.size = 30,
     vertex.label.color = "black",
     edge.color = "blue",
     main = "Researcher Collaboration Network(192324062")
```



Conclusion

The network contains **2 separate connected groups**:

Component 1: R1, R2, R3

Component 2: R4, R5, R6

Hence, **the network does not form one connected group; it has two connected components.**